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1 Hypothesis testing with maximum likelihood

In your experiments, some of you wrote a decision rule that chooses the hypothesis with the highest likelihood. Let me formalise this. You have some parameters θ_1 for hypothesis H_1 and some other parameters θ_0 for H_0. To make things unambiguous, the conditional probability of data D, under hypothesis H_i, with parameters θ_i will be written as $P_i(\cdot \mid \theta_i)$

You then find the maximum likelihood parameters for each case:

$$\theta_i^* = \arg\max_{\theta_i} P_i(D \mid \theta_i).$$

This simply means: find the parameter values that make the observed data most likely under hypothesis $\mathrm{H_i}$. This is exactly the maximum likelihood (ML) estimate you computed in the previous assignment. Note that the argmax returns the parameter values achieving the maximum likelihood, not the value of the likelihood itself (that is max).

Afterwards, you choose the hypothesis having the highest likelihood:

$$H* = \arg\max_{H_i} P_i(D \mid \theta_i^*).$$

For the same data, you will generally find that **more complex models have higher likelihood**.

1.1 Example

In the lung cancer example, you can have two simple hypotheses:

H_0: Smoking, cancer and asbestos are independent. Then $P_0(S, A, C) = P_0(S)P_0(A)P_0(C)$. These are all binary variables, so you need three parameters: $\theta_{0,S}, \theta_{0,C}, \theta_{0,A}$ for the probability of smoking, cancer and asbestos.

H_1: Smoking depends on cancer and asbestos exposure, but smoking and asbestos are independent. Then $P_1(S, A, C) = P_1(S)P_1(A)P_1(C|A, S)$. So you need $\theta_{1,S}, \theta_{1,A}, \theta_{1,C|AS}, \theta_{1,C|\neg AS}, \theta_{1,C|A\neg S}, \theta_{1,C|\neg A}$, i.e. the probability of cancer given smoking and asbestos has to be defined for all 4 combinations of S and A. So we have 6 parameters.

1.2 Questions

Now apply this thinking to your own problem. Make sure that H_0 is a strictly simpler model than H_1 (that is, the graph of H_1 has some added edges on top of H_0).

1. Write down the graphical model and factors for each model (if not already done).
2. List the variables you need to estimate in each case (if not already done)..
3. If you have **data generated from** H_0, how often do you choose H_1 using the maximum likelihood rule? How about generating data from H_1? How often do you choose H_0? Run the experiment for smaller (e.g. 10) and larger (e.g. 10'000) amounts of data. Average over 10 runs if you need to, but if H_1 is strictly more complex, it will always 'win'.
4. Does the same happen if you use MAP estimates to choose a hypothesis? Repeat the experiment.
5. We have already seen method for model selection in the beginning of the course. Could you use one of them to obtain a more robust method for choosing hypotheses? Run a small experiment to verify it.
6. Could you use the same approach when using e.g. parameter estimates from linear/logistic regression? Explain how and run a small experiment.

2 Discussion topics

For this submission, you are *encouraged* to submit your **Chatbot prompts and responses**, as well as topics of **group discussion**, relating to this week's *lecture*, *exercises* and the *assignment*. This will **not** be taken into account for your grade, but we would like to understand:

- which concepts you found difficult,
- which questions you asked,
- whether the responses were helpful, confusing, or incorrect,

- and what eventually helped you understand the material.

Feel free to include both successful and unsuccessful interactions with fellow students and chatbots alike.